

When does a box fit into a box?

An explicit criterion with a formal proof

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Abstract

We give a necessary and sufficient condition, in closed form, for a rectangular box with edges $q_1, q_2, q_3 \geq 0$ to fit, after an arbitrary rigid motion, into a rectangular box with edges $p_1, p_2, p_3 > 0$. The condition is a disjunction of nine explicit checks, each consisting of two planar tests expressed by a formula with a single square root. The key geometric fact is that if a box fits at all, it fits in a position where one of its edges is perpendicular to one of the axes of the container; we prove this by showing that at any other position all three widths of the box can be decreased simultaneously. The problem was posed by Garnett in 1923 and was described by Jerrard and Wetzel in 2004 as an instance of a general unsolved fitting problem for boxes in higher dimensions. The main theorem has been verified in the Lean 4 proof assistant with the Mathlib library, together with the analogous criterion for strict fitting (without touching the container). The main criterion is constructive: when the box fits, it yields an explicit rotation. An implementation decides fitting and produces a placement in about ten microseconds, three to four orders of magnitude faster than a multi-start local numerical search over rotations, and without its false negative answers near the boundary.

1 Introduction

Let $P = [0, p_1] \times [0, p_2] \times [0, p_3]$ with $p_i > 0$ and $Q = [0, q_1] \times [0, q_2] \times [0, q_3]$ with $q_j \geq 0$. We say that Q *fits into* P if $f(Q) \subseteq P$ for some isometry f of $\mathbb{R}^3$. The question of when this happens was posed by F. M. Garnett in 1923 [5]; W. B. Carver [2] gave a partial answer, assuming without proof that in an optimal position all vertices of Q lie on faces of P . The planar analogue, a rectangle in a rectangle, was settled by Carver in 1957 [3] (see also Wetzel [14]). Jerrard and Wetzel [6, p. 22] describe the three-dimensional question as an instance of a “general unsolved fitting problem for boxes in higher dimensions”. Among related results we mention the treatment of a disk in a box [1], of a circular cylinder along the space diagonal of a box [7], where a criterion “along the lines of Post” [11] is asked for and described as unsolved, and the containment algorithms of Martin and Stephenson [9]. The olympiad problem [8] proves the necessary condition $q_1 + q_2 + q_3 \leq p_1 + p_2 + p_3$.

In this paper we give such a criterion for boxes. It is a disjunction of explicit checks, much as Post’s criterion for a triangle in a triangle [11] is a disjunction of 18 inequalities.

The criterion

For $x \geq y \geq 0$ and $s \geq 0$ put

$$H(x, y; s) = \begin{cases} y, & x \leq s, \\ \min\left(x, \frac{2xys + (x^2 - y^2)\sqrt{x^2 + y^2 - s^2}}{x^2 + y^2}\right), & x > s, \end{cases} \quad (1)$$

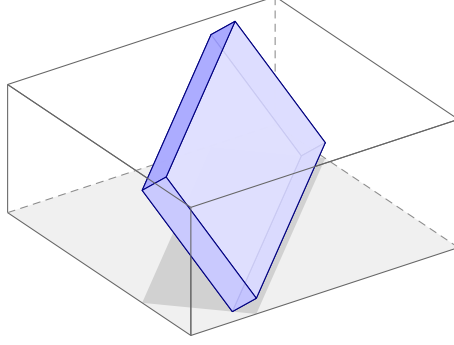


Figure 1: A box $2.8 \times 1.3 \times 0.25$ in a container of height 1.2 with floor 2.7×2.64 , placed as described by one of the checks of the criterion: the long edge is horizontal, the face 1.3×0.25 is tilted in the vertical strip, and the orthogonal projection of the box onto the floor (the shadow) is a $2.8 \times h^*$ rectangle rotated in the floor. The rotation is computed as in the proof of Proposition 4.1.

and for arbitrary $u, v \geq 0$ let $H(u, v; s) = H(\max(u, v), \min(u, v); s)$. By Lemma 3.1, when $\min(u, v) \leq s$ the number $H(u, v; s)$ is the least possible extent along a strip of width s of a $u \times v$ rectangle placed in the strip. For $a, b, u, v \geq 0$ put

$$\text{Fit}(a, b; u, v) :\iff \min(u, v) \leq a \text{ and } H(u, v; a) \leq b.$$

This says that the $u \times v$ rectangle fits into the $a \times b$ rectangle (Remark 3.2).

Definition 1.1 (Criterion NC). Let (i, i', i'') be a permutation of the axes $(1, 2, 3)$ of P and (j, k, l) a permutation of the edges $(1, 2, 3)$ of Q . The *check* $\text{NC}_{(i, i', i''), (j, k, l)}$ is

$$\min(q_k, q_l) \leq p_i \text{ and } \text{Fit}(p_{i'}, p_{i''}; q_j, h^*), \quad h^* := H(q_k, q_l; p_i).$$

$\text{NC}(p, q)$ holds if at least one check holds. Since H is symmetric in its first two arguments and $\text{Fit}(a, b; u, v)$ is equivalent to $\text{Fit}(b, a; u, v)$ (both say that the $u \times v$ rectangle fits into the $a \times b$ rectangle, Remark 3.2), a check depends only on the choice of (i, j) : there are 9 essentially different checks (this equivalence is also verified formally, Section 11).

Geometrically, the check describes positions in which edge j of Q is horizontal, i.e. perpendicular to axis i of P (the “height”). The face $q_k \times q_l$ then stands in the vertical strip of height p_i ; its orthogonal projection onto the floor is a segment of length at least h^* , and the rectangle $q_j \times h^*$ must fit into the floor $p_{i'} \times p_{i''}$ (Figure 1).

Theorem 1.2 (Main theorem). *Let $p_1, p_2, p_3 > 0$ and $q_1, q_2, q_3 \geq 0$. The box Q fits into the box P if and only if $\text{NC}(p, q)$ holds.*

Sufficiency is constructive (Section 4): every check that holds yields an explicit rotation. Necessity rests on the following theorem (Section 6), which is the main new ingredient. Here $w_i(R) = \sum_j q_j |R_{ij}|$ is the width of the rotated box along axis i (Section 2).

Theorem 1.3 (Descent). *Let $q_1, q_2, q_3 > 0$ and let $R \in \text{O}(3)$ have no zero entries. Then there is $R' \in \text{O}(3)$ with $w_i(R') < w_i(R)$ for $i = 1, 2, 3$.*

Formal verification

Theorem 1.2 has been formalized and checked in Lean 4 with Mathlib; see Section 11. The formal statement uses literally the formulas above; the only definitions on which its meaning depends are those of a box and of fitting.

2 Reduction to widths

Every isometry of $\mathbb{R}^3$ has the form $x \mapsto Rx + t$ with $R \in O(3)$ (Mazur–Ulam). Column j of R is the direction of edge j of the moved box, and R_{ij} is the cosine of the angle between this edge and axis i of P .

Lemma 2.1. *Let R be any real 3×3 matrix. Some translate of RQ lies in P if and only if*

$$w_i(R) := \sum_{j=1}^3 q_j |R_{ij}| \leq p_i \quad (i = 1, 2, 3).$$

Consequently Q fits into P if and only if $w(R) \leq p$ for some $R \in O(3)$.

Proof. The i -th coordinate of $Rx = \sum_j R_{ij}x_j$, $x \in Q$, ranges over an interval of length $\sum_j q_j |R_{ij}|$, the endpoints being attained at vertices of Q . The translations along different axes are independent. $\square$

The same argument in the plane shows that a $u \times v$ rectangle fits into an $a \times b$ rectangle if and only if $uc + vd \leq a$ and $ud + vc \leq b$ for some $c, d \geq 0$ with $c^2 + d^2 = 1$.

3 A rectangle in a strip

Let $x \geq y \geq 0$ and $s \geq 0$. For $c, d \geq 0$ with $c^2 + d^2 = 1$ put $A = xc + yd$ (the extent of the rotated $x \times y$ rectangle across the strip) and $B = xd + yc$ (its extent along the strip).

Lemma 3.1 (Strip lemma). *(a) If $y \leq s$, there are $c, d \geq 0$ with $c^2 + d^2 = 1$, $A \leq s$ and $B \leq H(x, y; s)$.*

(b) If $c, d \geq 0$, $c^2 + d^2 = 1$ and $A \leq s$, then $y \leq s$ and $H(x, y; s) \leq B$.

The same holds for arbitrary $u, v \geq 0$ in place of x, y with $H(u, v; s)$, $A = uc + vd$, $B = ud + vc$ (exchange c and d if $u < v$).

Proof. If $x = 0$ everything is trivial, so let $r = \sqrt{x^2 + y^2} > 0$ and let $\alpha \in [0, \pi/4]$ be given by $\cos \alpha = x/r$, $\sin \alpha = y/r$. Write $(c, d) = (\cos \theta, \sin \theta)$ with $\theta \in [0, \pi/2]$. Then

$$A(\theta) = r \cos(\theta - \alpha), \quad B(\theta) = r \sin(\theta + \alpha).$$

For $\theta \in [0, \pi/2]$ we have $\theta - \alpha \in [-\pi/2, \pi/2]$ and $\theta + \alpha \in [0, \pi]$, so both A and B are concave on $[0, \pi/2]$; hence on every subinterval they attain their minima at the endpoints. Moreover $A(0) = B(\pi/2) = x$ and $A(\pi/2) = B(0) = y$.

By concavity $A \geq \min(x, y) = y$, which proves the first claim of (b). If $x \leq s$, the angle $\theta = 0$ is admissible with $B(0) = y$, and $B \geq \min(B(0), B(\pi/2)) = y$ everywhere; this proves both (a) and (b).

Let $y \leq s < x$. On $[0, \alpha]$ the function A increases from x to r , so $A > s$ there; on $[\alpha, \pi/2]$ it decreases from r to $y \leq s$. Hence the admissible angles form an interval $[\theta^*, \pi/2]$ with $A(\theta^*) = s$, and the minimum of the concave function B over it equals $\min(B(\theta^*), B(\pi/2)) = \min(B(\theta^*), x)$. It remains to compute $B(\theta^*)$. With $\varphi = \theta^* - \alpha \in [0, \pi/2]$ we have $\cos \varphi = s/r$, $\sin \varphi = \sqrt{r^2 - s^2}/r$, and

$$\cos \theta^* = \frac{xs - y\sqrt{r^2 - s^2}}{r^2}, \quad \sin \theta^* = \frac{ys + x\sqrt{r^2 - s^2}}{r^2}, \quad B(\theta^*) = \frac{2xys + (x^2 - y^2)\sqrt{r^2 - s^2}}{r^2}.$$

Thus $\min\{B : A \leq s\} = H(x, y; s)$, which gives (a) (take the minimizing endpoint) and (b). $\square$

Remark 3.2. By the planar version of Lemma 2.1 and Lemma 3.1, $\text{Fit}(a, b; u, v)$ holds if and only if the $u \times v$ rectangle fits into the $a \times b$ rectangle. This is Carver's condition [3]: for $u \geq v$, $a \geq b$ his criterion reads “ $u \leq a$ and $v \leq b$, or $u > a$ and $b \geq (2uva + (u^2 - v^2)\sqrt{u^2 + v^2 - a^2})/(u^2 + v^2)$ ”, and for $u > a$ the last inequality is exactly $H(u, v; a) \leq b$. Our form differs only in that the sides need not be ordered. Wetzel [14, Theorem 1] gives the equivalent form $(\frac{a+b}{u+v})^2 + (\frac{a-b}{u-v})^2 \geq 2$ (for $u > a$, $v \leq b$), which goes back to Dunkel's equation of an inscribed rectangle [4]. (Numerically Fit and Wetzel's form agree on $2 \cdot 10^5$ random rectangles; see `tools/box_fit.py`.)

4 Sufficiency

Proposition 4.1. *If a check of $\text{NC}(p, q)$ holds, then Q fits into P , and an explicit rotation can be written down.*

Proof. Relabeling axes and edges (which permutes rows and columns of R) we may assume that the check is $\text{NC}_{(1,2,3),(1,2,3)}$: $\min(q_2, q_3) \leq p_1$ and $\text{Fit}(p_2, p_3; q_1, h^*)$ with $h^* = H(q_2, q_3; p_1)$. By Lemma 3.1(a) there are $c_1, d_1 \geq 0$, $c_1^2 + d_1^2 = 1$, with

$$q_2 c_1 + q_3 d_1 \leq p_1, \quad q_2 d_1 + q_3 c_1 \leq h^*,$$

in particular $h^* \geq 0$. Applying Lemma 3.1(a) to the rectangle $q_1 \times h^*$ in the strip of width p_2 gives $c_2, d_2 \geq 0$, $c_2^2 + d_2^2 = 1$, with

$$q_1 c_2 + h^* d_2 \leq p_2, \quad q_1 d_2 + h^* c_2 \leq H(q_1, h^*; p_2) \leq p_3.$$

Let

$$R = \begin{pmatrix} 0 & c_1 & -d_1 \\ c_2 & -d_1 d_2 & -c_1 d_2 \\ d_2 & d_1 c_2 & c_1 c_2 \end{pmatrix}.$$

Its columns $f_1 = (0, c_2, d_2)$, $f_2 = c_1 e_1 + d_1 n$, $f_3 = -d_1 e_1 + c_1 n$ with $n = (0, -d_2, c_2)$ are orthonormal, so $R \in \text{O}(3)$. Its widths are

$$\begin{aligned} w_1 &= q_2 c_1 + q_3 d_1 \leq p_1, \\ w_2 &= q_1 c_2 + d_2(q_2 d_1 + q_3 c_1) \leq q_1 c_2 + d_2 h^* \leq p_2, \\ w_3 &= q_1 d_2 + c_2(q_2 d_1 + q_3 c_1) \leq q_1 d_2 + c_2 h^* \leq p_3, \end{aligned}$$

and Lemma 2.1 concludes the proof. $\square$

5 Positions with a horizontal edge

Proposition 5.1. *Let $q \geq 0$, $p > 0$, and let $R \in \text{O}(3)$ satisfy $w(R) \leq p$ and $R_{ij} = 0$ for some i, j . Then $\text{NC}(p, q)$ holds.*

Proof. Relabeling, let $R_{11} = 0$. For an orthogonal matrix $R = (\det R) \text{cof}(R)$ and $\det R = \pm 1$, so each $|R_{ab}|$ equals the absolute value of the corresponding cofactor. Since $R_{11} = 0$, this gives

$$|R_{22}| = |R_{13}||R_{31}|, \quad |R_{23}| = |R_{12}||R_{31}|, \quad |R_{32}| = |R_{13}||R_{21}|, \quad |R_{33}| = |R_{12}||R_{21}|.$$

Put $c = |R_{12}|$, $d = |R_{13}|$, $c' = |R_{21}|$, $d' = |R_{31}|$; then $c^2 + d^2 = 1$ (first row) and $c'^2 + d'^2 = 1$ (first column). With $B := q_2 d + q_3 c$ we obtain

$$w_1 = q_2 c + q_3 d, \quad w_2 = q_1 c' + d' B, \quad w_3 = q_1 d' + c' B.$$

From $w_1 \leq p_1$ and Lemma 3.1(b), $\min(q_2, q_3) \leq p_1$ and $h^* := H(q_2, q_3; p_1) \leq B$; also $h^* \geq 0$ by Lemma 3.1(a). Hence $q_1 c' + d' h^* \leq w_2 \leq p_2$ and $q_1 d' + c' h^* \leq w_3 \leq p_3$. Lemma 3.1(b), applied to the rectangle $q_1 \times h^*$ in the strip of width p_2 with direction (c', d') , gives $\min(q_1, h^*) \leq p_2$ and $H(q_1, h^*; p_2) \leq q_1 d' + h^* c' \leq p_3$, i.e. $\text{Fit}(p_2, p_3; q_1, h^*)$. Thus $\text{NC}_{(1,2,3),(1,2,3)}$ holds. $\square$

6 Descent: proof of Theorem 1.3

Throughout this section $q_1, q_2, q_3 > 0$ and $R \in \text{O}(3)$ has no zero entries. We write e_1, e_2, e_3 for the standard basis, $u \cdot v$ for the scalar product and $u \times v$ for the vector product.

6.1 Linearization

Let $s_{ij} = \text{sgn } R_{ij} \in \{\pm 1\}$, $g_i = (s_{i1}q_1, s_{i2}q_2, s_{i3}q_3)$ and $a_i = Rg_i \in \mathbb{R}^3$. Write $A_{ik} = (a_i)_k$. Then $A_{ii} = \sum_j R_{ij}s_{ij}q_j = w_i(R)$. If $C \in \text{O}(3)$ and CR has the same sign pattern as R , then

$$w_i(CR) = \sum_j q_j s_{ij} (CR)_{ij} = (Ca_i)_i. \quad (2)$$

Lemma 6.1. $|A_{ik}| < A_{kk}$ for $i \neq k$.

Proof. $|A_{ik}| = |\sum_j s_{ij}q_j R_{kj}| \leq \sum_j q_j |R_{kj}| = A_{kk}$. Equality would force all $s_{ij}R_{kj}$ ($q_j > 0$, $R_{kj} \neq 0$) to have the same sign, hence all products $R_{ij}R_{kj}$ to be nonzero of the same sign, contradicting the orthogonality of rows i and k . $\square$

6.2 The Cayley rotation

For $v \in \mathbb{R}^3$ let $[v]_\times x = v \times x$ and

$$C(v) = \frac{(1 - |v|^2)I + 2[v]_\times + 2vv^\top}{1 + |v|^2}.$$

This is the rotation given by the quaternion $(1, v)$; one checks directly that $C(v)C(v)^\top = I$. Moreover $C(0) = I$ and C is continuous. For $a \in \mathbb{R}^3$ put

$$b_i(a) = a \times e_i, \quad P_i(a; v) = v_i(v \cdot a) - |v|^2 a_i, \quad D_i(a; v) = b_i(a) \cdot v + P_i(a; v).$$

Since $e_i \cdot (v \times a) = v \cdot (a \times e_i)$, a direct computation gives

$$(C(v)a)_i - a_i = \frac{2}{1 + |v|^2} D_i(a; v). \quad (3)$$

Note that $P_i(a; \cdot)$ is a quadratic form: $P_i(a; tv) = t^2 P_i(a; v)$. This is why we use the Cayley parametrization: by (3) the change of each width is, up to a positive factor, an explicit polynomial of degree two in v , so both its first- and second-order terms are available.

6.3 Descent along a curve

We shall rotate along $v(t) = t(h + tz)$, $t > 0$ small. Then

$$D_i(a; v(t)) = t b_i(a) \cdot h + t^2 (b_i(a) \cdot z + P_i(a; h + tz)). \quad (4)$$

Lemma 6.2. *Let $a, h, z \in \mathbb{R}^3$ and $b = b_i(a)$.*

- (a) *If $b \cdot h < 0$, or $b \cdot h = 0$ and $b \cdot z + P_i(a; h) < 0$, then $D_i(a; v(t)) < 0$ for all sufficiently small $t > 0$.*
- (b) *If $b = 0$, $a_i > 0$ and all components of z are nonzero, then $D_i(a; v(t)) < 0$ for all sufficiently small $t > 0$.*

Proof. (a) By (4), $D_i/t = b \cdot h + t(b \cdot z + P_i(a; h + tz)) \rightarrow b \cdot h$ as $t \rightarrow 0^+$; in the second case $D_i/t^2 = b \cdot z + P_i(a; h + tz) \rightarrow b \cdot z + P_i(a; h)$. (b) $b = a \times e_i = 0$ means $a = a_i e_i$; then $P_i(a; u) = u_i^2 a_i - |u|^2 a_i = -a_i \sum_{m \neq i} u_m^2$. For small $t > 0$ every component $h_m + tz_m$ of $u = h + tz$ is nonzero (if $h_m \neq 0$ by continuity, if $h_m = 0$ because $z_m \neq 0$). Hence $D_i(a; v(t)) = t^2 P_i(a; h + tz) < 0$. $\square$

6.4 The key inequality

Lemma 6.3. *Let $a_1, a_2, a_3 \in \mathbb{R}^3$ satisfy $|A_{ik}| < A_{kk}$ for $i \neq k$, where $A_{ik} = (a_i)_k$. Let $\nu \in \mathbb{R}^3$, $\nu \geq 0$, have at least two positive entries and satisfy $\sum_i \nu_i b_i(a_i) = 0$. Then $\sum_i \nu_i P_i(a_i; h) < 0$ for every $h \neq 0$.*

Proof. Since $a \times e_1 = (0, a_3, -a_2)$, $a \times e_2 = (-a_3, 0, a_1)$, $a \times e_3 = (a_2, -a_1, 0)$, the condition $\sum_i \nu_i b_i(a_i) = 0$ says precisely that

$$\nu_i A_{ik} = \nu_k A_{ki} \quad (i \neq k).$$

Put $d_i = \nu_i A_{ii} \geq 0$ and $m_{ik} = \nu_i A_{ik} = \nu_k A_{ki}$. Expanding and using the symmetry,

$$\sum_i \nu_i P_i(a_i; h) = \sum_i \nu_i \left(h_i \sum_k h_k A_{ik} - |h|^2 A_{ii} \right) = - \sum_{i < k} B_{ik}, \quad B_{ik} = d_i h_k^2 + d_k h_i^2 - 2m_{ik} h_i h_k.$$

By the hypothesis, $m_{ik}^2 = \nu_i \nu_k A_{ik} A_{ki} \leq \nu_i \nu_k |A_{ik}| |A_{ki}| \leq \nu_i \nu_k A_{kk} A_{ii} = d_i d_k$, with strict inequality if $\nu_i \nu_k > 0$.

Each $B_{ik} \geq 0$: if $d_i = 0$ then $m_{ik} = 0$ and $B_{ik} = d_k h_i^2$; if $d_i > 0$ then $d_i B_{ik} = (d_i h_k - m_{ik} h_i)^2 + (d_i d_k - m_{ik}^2) h_i^2 \geq 0$.

Some B_{ik} is positive. Choose c with $h_c \neq 0$ and, using the two positive entries of ν , an index $i \neq c$ with $\nu_i > 0$, so $d_i > 0$. If $d_c = 0$ then $m_{ic} = 0$ and $B_{ic} = d_i h_c^2 > 0$. If $d_c > 0$ then $\nu_c > 0$, hence $m_{ic}^2 < d_i d_c$, and $d_i B_{ic} = (d_i h_c - m_{ic} h_i)^2 + (d_i d_c - m_{ic}^2) h_i^2$ is positive both for $h_i \neq 0$ and for $h_i = 0$. $\square$

6.5 Choice of the curve

The idea is as follows. Either some direction h decreases all widths to first order, or the first-order terms $b_i \cdot h$ are blocked by a nonnegative relation $\sum_i \nu_i b_i = 0$; in the latter case Lemma 6.3 shows that the second-order terms decrease the widths. We use Motzkin's theorem of the alternative [10, 12] in the following form: for vectors $b_i \in \mathbb{R}^3$ and numbers c_i , $i \in K$, either the system $b_i \cdot z < c_i$ ($i \in K$) has a solution, or there is $\mu \geq 0$, $\mu \neq 0$, supported on K , with $\sum_i \mu_i b_i = 0$ and $\sum_i \mu_i c_i \leq 0$.

Lemma 6.4. *Let a_i be as in the Linearization, $b_i = b_i(a_i)$ and $K = \{i : b_i \neq 0\}$. There are $h, z \in \mathbb{R}^3$ such that all components of z are nonzero and, for each $i \in K$, either $b_i \cdot h < 0$, or $b_i \cdot h = 0$ and $b_i \cdot z + P_i(a_i; h) < 0$.*

Proof. Apply Motzkin's theorem with $c = 0$. If $b_i \cdot h < 0$ for all $i \in K$ has a solution h , take $z = (1, 1, 1)$. Otherwise there is $\mu \neq 0$, $\mu \geq 0$, with $\sum \mu_i b_i = 0$ (note $b_i = 0$ for $i \notin K$), so the matrix with rows b_1, b_2, b_3 is singular and there is $h \neq 0$ with $b_i \cdot h = 0$ for all i . Apply Motzkin's theorem again, with $c_i = -P_i(a_i; h)$.

If there is z_0 with $b_i \cdot z_0 + P_i(a_i; h) < 0$ for $i \in K$, then $z = z_0 + \delta(1, 1, 1)$ with small $\delta > 0$ keeps these strict inequalities and has all components nonzero.

Otherwise there is $\nu \geq 0$, $\nu \neq 0$, supported on K , with $\sum \nu_i b_i = 0$ and $\sum \nu_i P_i(a_i; h) \geq 0$. If ν had a single positive entry ν_i , then $\nu_i b_i = 0$ would give $b_i = 0$, contradicting $i \in K$. So ν has two positive entries, and Lemma 6.3 (whose hypothesis holds by Lemma 6.1) gives $\sum \nu_i P_i(a_i; h) < 0$, a contradiction. $\square$

6.6 Proof of Theorem 1.3

Take h, z from Lemma 6.4 and put $R(t) = C(v(t))R$. For all sufficiently small $t > 0$:

- $R(t) \in O(3)$ has the same sign pattern as R , because $R(t) \rightarrow R$ and R has no zero entries;
- $D_i(a_i; v(t)) < 0$ for each i : for $i \in K$ by Lemma 6.2(a), and for $i \notin K$ by Lemma 6.2(b), since $(a_i)_i = w_i(R) > 0$.

By (2) and (3), $w_i(R(t)) - w_i(R) = (C(v(t))a_i)_i - (a_i)_i = \frac{2}{1+|v(t)|^2} D_i(a_i; v(t)) < 0$ for $i = 1, 2, 3$. $\square$

7 Proof of the main theorem

Sufficiency is Proposition 4.1. For necessity, by Proposition 5.1 it suffices to show: *if Q fits into P , then $w(R) \leq p$ for some $R \in O(3)$ with a zero entry.*

Case $q_1, q_2, q_3 > 0$. The function $F(R) = \max_i w_i(R)/p_i$ is continuous on the compact group $O(3)$ and attains its minimum at some R^* ; since Q fits, $F(R^*) \leq 1$, i.e. $w(R^*) \leq p$. If R^* had no zero entries, Theorem 1.3 would give R' with $w_i(R') < w_i(R^*)$ for all i , hence $F(R') < F(R^*)$. So R^* has a zero entry.

General case $q \geq 0$. Let $w(R_0) \leq p$ and suppose that no R in the compact set $Z = \{R \in O(3) : R_{ij} = 0 \text{ for some } i, j\}$ satisfies $w(R) \leq p$. Then the continuous function $\varphi(R) = \max_i (w_i(R) - p_i)$ is positive on Z and has a minimum $m > 0$ there. Let $\delta = m/6$, $q' = q + \delta(1, 1, 1) > 0$ and $p' = p + 3\delta(1, 1, 1)$. Since $\sum_j |R_{ij}| \leq 3$ for $R \in O(3)$, we have $w_i(R_0; q') = w_i(R_0; q) + \delta \sum_j |(R_0)_{ij}| \leq p'_i$, so q' fits into p' . By the previous case there is $R_1 \in Z$ with $w(R_1; q') \leq p'$, hence $w_i(R_1; q) \leq w_i(R_1; q') \leq p_i + 3\delta$ and $\varphi(R_1) \leq 3\delta < m$, a contradiction. $\square$

8 Examples

Example 8.1 (A rod). Let $q = (L, 0, 0)$. Take the check with height $i = 1$, horizontal edge $j = 2$ and $(k, l) = (1, 3)$. Then $h^* = H(L, 0; p_1) = \sqrt{\max(L^2 - p_1^2, 0)}$, and $\text{Fit}(p_2, p_3; 0, h^*)$ reduces to $h^{*2} \leq p_2^2 + p_3^2$. Thus the check holds if and only if $L^2 \leq p_1^2 + p_2^2 + p_3^2$; conversely, a longer rod cannot fit since $\sqrt{p_1^2 + p_2^2 + p_3^2}$ is the diameter of P . The rod fits exactly along the space diagonal, and this position has an edge (of length 0) perpendicular to an axis.

Example 8.2 (A square in a cube). The largest square that fits into the unit cube has side $a = 3\sqrt{2}/4$ (Nieuwland, published by van Swinden [13]). For $q = (a, a, 0)$ and $p = (1, 1, 1)$ take $j = 1$, $(k, l) = (2, 3)$: $h^* = H(a, 0; 1) = \sqrt{a^2 - 1} = \sqrt{2}/4$. For the floor, $x^2 = 9/8$, $y^2 = 1/8$, $x^2 + y^2 = 5/4$, $\sqrt{x^2 + y^2 - 1} = 1/2$, $2xy = 3/4$, so the second branch of (1) equals $(3/4 + 1 \cdot \frac{1}{2})/(5/4) = 1$ and $H(a, h^*; 1) = \min(a, 1) = 1$: the check holds with equality, in accordance with Nieuwland's result.

Example 8.3 (Rectangle in a cube). Jerrard and Wetzel [6, Theorem 5] found the largest $L \times \lambda L$ rectangle ($0 < \lambda \leq 1$) that fits into the unit cube: $L_{\max} = 3/(\sqrt{3 - \lambda^2} + \sqrt{2}\lambda)$ for $\lambda < 1 - \sqrt{2}/2$, $L_{\max} = \sqrt{2}$ for $1 - \sqrt{2}/2 \leq \lambda \leq \sqrt{2}/2$, and $L_{\max} = 3/(\sqrt{2} + \sqrt{3\lambda^2 - 1})$ for $\lambda > \sqrt{2}/2$ (for $\lambda = 1$ this is Nieuwland’s square). This is a special case of Theorem 1.2: $q = (L, \lambda L, 0)$, $p = (1, 1, 1)$. The largest L for which NC holds agrees with this formula: on 2000 random values of λ the discrepancy is at most $5 \cdot 10^{-16}$ (a numerical, not a formal, check).

Example 8.4 (Classical necessary conditions are not sufficient). For $p = (1, 1, 1)$ and $q = (1.1, 1.1, 0.01)$ the sums of edges, of squared edges, the surface areas and the volumes are all smaller for Q , yet Q does not fit: its 1.1×1.1 face would give a square larger than $3\sqrt{2}/4$ in the cube.

9 Computation

The Lean definition, and the function `nc` of the reference implementation, run through all 36 pairs of permutations; only 9 checks are essentially different (Definition 1.1), and the function `nc_fast` performs only these. Each check takes two evaluations of formula (1) and a few comparisons, so the criterion takes constant time. When a check holds, the proof of Proposition 4.1 gives the rotation explicitly: two angles from Lemma 3.1(a), one for the face in the vertical strip and one for the rectangle on the floor. The reference implementation `tools/box_fit.py` in the repository (Section 11) returns a rotation matrix R with $\det R = 1$ and a translation t such that $x \mapsto Rx + t$ maps Q into P ; its self-test verifies about $2 \cdot 10^4$ random placements (orthogonality of R and all vertices inside P).

For comparison we used a numerical search that minimises $\max_i w_i(R)/p_i$ over rotations by the Nelder–Mead method from random starts and answers “fits” when some start reaches a value ≤ 1 . On 200 random pairs near the boundary (the box scaled by a factor $1 \pm e$ from the largest fitting size, $10^{-4} \leq e \leq 10^{-2}$; averages over several runs, Python, one core of a desktop computer) the criterion takes 34 microseconds per pair with 36 checks and 9 microseconds with 9 checks, and the explicit placement 14 microseconds, while the numerical search takes 78, 234 and 554 milliseconds per pair with 5, 20 and 50 starts, i.e. it is $2 \cdot 10^3$ to $1.6 \cdot 10^4$ times slower even than the 36-check version, and gives on average 26, 4 and 0.3 wrong answers out of 200. All its errors are “does not fit” for boxes that fit: a local search can confirm that a box fits, but its failure to find a position proves nothing. A rigorous negative answer requires a global method with guaranteed bounds, such as a branch-and-bound search over rotations, which is considerably more expensive; the criterion gives both answers exactly. The script is `tools/benchmark.py`.

10 Strict fitting

Remark 10.1. Say that Q fits strictly into P if $f(Q)$ lies in the open box $(0, p_1) \times (0, p_2) \times (0, p_3)$ for some isometry f , i.e. without touching the boundary of P . Let H° be defined as H in (1), but with the case split $x < s / x \geq s$ instead of $x \leq s / x > s$, and let NC° be the criterion of Definition 1.1 in which all inequalities are strict and H is replaced by H° . Then, for $p > 0$ and $q \geq 0$:

- Q fits strictly into P if and only if $\text{NC}(p - \varepsilon(1, 1, 1), q)$ holds for some $0 < \varepsilon < \min_i p_i$;
- Q fits strictly into P if and only if $\text{NC}^\circ(p, q)$ holds.

Complete formal proofs are in `lean/Boxes/Strict.lean` (theorems `fitsStrict_iff_shrink` and `fitsStrict_iff_NC`); they follow Sections 2–7 with strict inequalities.

The change of the case split is essential, because $H(x, y; s)$ jumps at $s = x$: a face whose longer side equals the height may stand upright touching the ceiling, but if touching is forbidden it cannot be tilted slightly (a small tilt increases its height). For example, let $p = (1, 1, 1)$ and $q = (1, 0.5, 0.5)$. Making strict only the inequalities of the checks, with H unchanged, gives a check that holds: with a horizontal edge 0.5, $h^* = H(1, 0.5; 1) = 0.5$ and 0.5×0.5 fits strictly into 1×1 . But NC° fails: here $H^\circ(1, 0.5; 1) = \min(1, 0.5 \cdot 2.75/1.25) = 1$, and each check requires $H^\circ(1, 0.5; 1) < 1$ (for the floor if the horizontal edge is 1, and for h^* and then the floor if it is 0.5). So this box does not fit strictly.

11 Formal verification

The accompanying repository <https://github.com/MikhailLevit1987/box-in-box> (archived at Zenodo, [doi:10.5281/zenodo.22974798](https://doi.org/10.5281/zenodo.22974798)) contains a Lean 4 formalization (Lean v4.34.0-rc2, Mathlib) of Theorem 1.2 and of the statements of Remark 10.1, about 1900 lines in eight files:

- `Reduction.lean`, `Statement.lean`: definitions, Lemma 2.1 (via Mazur–Ulam), Lemma 3.1(a), Proposition 4.1;
- `ZeroEntry.lean`: Lemma 3.1(b) (by a chord argument instead of concavity), Proposition 5.1;
- `Motzkin.lean`: the theorem of the alternative (via Hahn–Banach separation);
- `Descent.lean`: Section 6;
- `Necessity.lean`: Section 7 and the main theorem `fits_iff_NC`;
- `Strict.lean`: Section 10, theorems `fitsStrict_iff_shrink` and `fitsStrict_iff_NC`s (with two further definitions, the open box and strict fitting);
- `NineChecks.lean`: the criterion NC9 with the 9 essentially different checks and the theorem `NC_iff_NC9` stating that it is equivalent to NC.

The statement and the two definitions on which its meaning depends read verbatim:

```
def box (q : Fin 3 → ℝ) : Set E3 := {x | ∀ i, 0 ≤ x i ∧ x i ≤ q i}
def Fits (p q : Fin 3 → ℝ) : Prop := ∃ f : E3 ≃_i E3, f '' box q ⊆ box p
theorem fits_iff_NC (p q : Fin 3 → ℝ) (hp : ∀ i, 0 < p i) (hq : ∀ i, 0 ≤ q i) :
  Fits p q ↔ NC p q
```

Here $E3$ is the Euclidean space $\mathbb{R}^3$, `box q` is the set of points x with $0 \leq x_i \leq q_i$, and `Fits p q` means that some isometry f of $\mathbb{R}^3$ maps `box q` into `box p`. The criterion NC is Definition 1.1 verbatim (with indices 0, 1, 2). The command `#print axioms Boxes.fits_iff_NC` reports only the standard axioms `propext`, `Classical.choice`, `Quot.sound`.

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